

1 Explainability in Reinforcement Learning

2 Maxime Alaarabiou

3 Sunday 6th September, 2026

Chapter 1

Explainability in Reinforcement Learning

Today, the majority of Artificial Intelligence (AI) agents with which humans interact are deployed in controlled or low-risk environments. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences [Silver et al., 2017]. Even more autonomous systems, such as agents based on Large Language Model (LLM), remain constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval [Chen et al., 2021, Lewis et al., 2021]. Safety-critical applications integrating machine learning components are still rare. Introducing autonomous agents in such environments, requires the establishment of strong guarantees [Seshia et al., 2022]. These requirements include in particular:

- validating the system’s functioning before deployment,
- identifying the origin of a malfunction in case of failure,
- and certifying that an observed failure will not occur again.

The main challenge lies in the fact that the performing agents currently available rely on both Reinforcement Learning (RL) and Deep Learning (DL) [Mnih et al., 2013]. As a result, the mechanisms underlying their decisions cannot be understood directly from the learned model. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, the field of eXplainable Reinforcement Learning (XRL) has emerged as a subfield of eXplainable Artificial Intelligence (XAI). The objective of XRL is to develop explainability methods that make it possible to obtain explanations of RL agents. Due to the relevance of the topic and to foster broader real-world adoption of RL, a growing number of research teams are working on developing new explainability methods. This enthusiasm is reflected in a rapid increase in the number of publications, as illustrated in Figure 1.1.¹

¹The complete search protocol, including the databases consulted and the full set of query terms, has been archived on Zenodo <https://zenodo.org/records/19387785>.

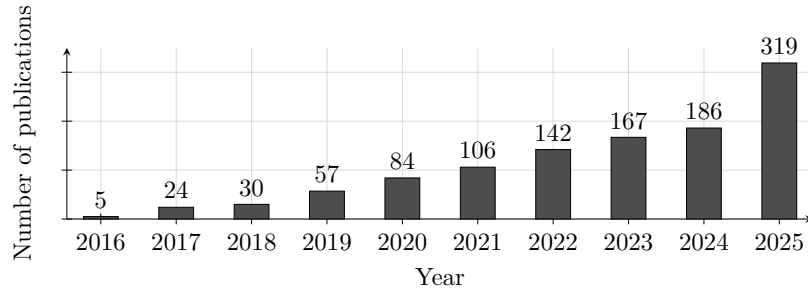


Figure 1.1: Yearly growth of Explainable Reinforcement Learning research with at least one citation on arXiv.

The growing number and diversity of these methods make it increasingly difficult to obtain a unified view of the field. This motivated our work on developing a unified taxonomy of explainability methods for RL, initially presented in [Alaarabiou et al., 2024]. The taxonomy provides a common structure for organizing existing approaches according to the type of information they use and the form of explanation they provide. Building on this work, this chapter provides a detailed presentation of the proposed taxonomy and uses it to organize a broader state-of-the-art review of XRL.

This chapter is organized as follows:

- Section 1.1 begins by clarifying what we mean by an explanation in the context of XAI.
- Section 1.1 then introduces a taxonomy for categorizing existing XRL methods.
- Section 1.3 provides a comprehensive state-of-the-art review structured around this taxonomy.
- Section 1.4 concludes the chapter with a summary and discussion.

1.1 Explanation in Artificial Intelligence

In this section, we define what an explanation is in the context of XAI and how it can be obtained. We first introduce the concept of explanation in general. We then describe how this concept applies to XAI. Finally, we further detail the properties that characterize an explainability method in the context of XAI.

1.1.1 Explanation

Defining what constitutes an explanation is challenging because the concept is inherently multidisciplinary. When we refer to explanation, we are inherently referring to an explanation provided to a human. As soon as humans are involved, the problem is no longer purely computational or mathematical. It involves several fields of research, such as philosophy, psychology, and sociology. So far, these disciplines have not converged on a unified definition of explanation that can be directly applied to XAI [Miller, 2019, Bellucci et al., 2021].

For this reason, we propose the following definition of explanation. An explanation involves two entities: the object (Definition 1.1.1) being explained and the receiver of the explanation. The receiver is a human who aims to understand how the object operates and will be referred to as the observer throughout this manuscript (Definition 1.1.2). The observer seeks to understand the object and, in doing so, builds a mental model of its functioning. An explanation is therefore what allows the observer to refine and correct their mental model of how the object operates (Definition 1.1.4). This refinement is obtained through a transformation process that produces an explanation from the object being explained. We refer to this process as an explainability method (Definition 1.1.5), detailed further in Section 1.1.3. For an observer to understand an explanation, the information it contains must be interpretable (Definition 1.1.3). Information is considered interpretable when it can be directly understood by a human without requiring any additional transformation process.

Definition 1.1.1: Object

The object of an explanation is the system the observer wants to understand.

Definition 1.1.2: Observer

The observer is the human who aims to understand an object. The observer possesses a mental model of the object being explained. This mental model is an internal representation of how the observer believes the system operates. This representation may be incomplete or inaccurate.

Definition 1.1.3: Interpretable

Information is interpretable when it can be directly understood by an observer without requiring any additional transformation process.

Definition 1.1.4: Explanation

An explanation is any interpretable information that enables an observer to refine or correct their mental model of the object of the explanation.

Definition 1.1.5: Explainability Method

An explainability method is a transformation process that produces an explanation from the object being explained.

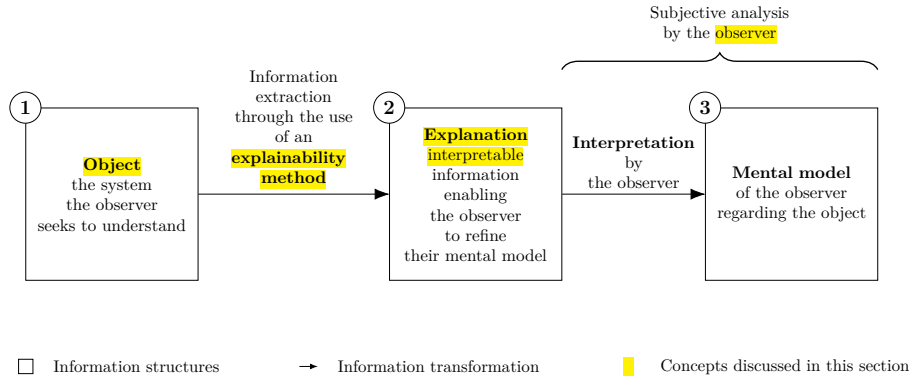


Figure 1.2: Explanation process.

Figure 1.2 summarizes this whole explanation process, from the object of the explanation to the mental model refined by the observer, as a chain of three information structures, depicted as boxes and connected by arrows denoting a transformation of information:

- (1) The object of the explanation, that is, the system the observer seeks to understand.
- (2) The explanation, the interpretable information produced from the object. This transformation is performed through the use of an explainability method.
- (3) The mental model held by the observer regarding the object.

A brace spanning boxes (2) and (3) indicates that this second half of the process, from the explanation to the resulting mental model, constitutes a subjective analysis carried out by the observer.

This definition of explanation establishes the approach to explainability adopted throughout this manuscript. Under this definition, explainability is regarded as a continuum that evolves as knowledge develops [Siegel and Craver, 2024, Zednik, 2019].

This quantitative perspective highlights that not all explanations provide the same level of value [Doshi-Velez and Kim, 2017]. The effectiveness of an explanation depends on its ability to improve the observer's understanding of the system. For example, describing a marginal aspect of a system is different from correcting an observer's misunderstanding of a key mechanism.

1.1.2 Explainable Artificial Intelligence

Now that the general framework of explanation has been established, we can examine how this concept applies in the context of AI. In the context of XAI, the object of explanation is an AI system. More precisely, XAI emerged as a response to the success of DL models [Gunning and Aha, 2019, Barredo Arrieta et al., 2020]. In these architectures, information is represented in a distributed manner across the network through patterns of activation and connections between computational units. Unlike symbolic approaches, individual components of Neural Network (NN) cannot be directly associated with human concepts. From this starting point, we identify two types of approaches:

- **Substitutive explainability:** In this approach, the assumption is that there is nothing to explain within a NN. There is no semantic meaning associated with each computational component, and therefore nothing to interpret. This corresponds to the “black box” view of NN. Under this assumption, XAI consists of replacing as much of the DL system as possible with alternative AI methods [Rudin, 2019].
- **Intrinsic explainability:** In this approach, it is argued that NN contain mechanisms that can be explained. The limitation is considered to lie not in the networks themselves, but in the absence of adequate tools to analyze their functioning [Olah et al., 2020].

This manuscript considers methods from both approaches in order to provide a comprehensive overview of the XAI field. Although both approaches have produced convincing results, this manuscript primarily adopts the perspective of the *Mechanistic Hypothesis* (Hyp. 1) [Siegel and Craver, 2024, Zednik, 2019]. This perspective is more closely aligned with the definition of explainability adopted in this manuscript, which considers explainability as a continuum that evolves with the development of knowledge.

Hypothesis 1: Mechanistic Hypothesis

Every AI system results from a sequence of causal computations. Because the rules governing these interactions are mechanical, the behavior of the system can be explained. If an AI system cannot be explained, this reflects a limitation of knowledge rather than an intrinsic property of the system.

1.1.3 Explainability Method in XAI

Now that the general framework of XAI has been established, we need to further detail the properties of an explanation in this context. In our framework, an explanation in the field of XAI can be characterized by three components:

- **Source:** refers to the origin of the information used by the explainability method. The explainability method transforms the raw information produced by the AI system into an explanation that can be understood by an observer.
- **Form:** refers to the representation through which the explanation is communicated. The form defines how the information is presented to the observer and must allow human interpretation. An explanation can take many different forms, including text, images, computer code, and mathematical equations.
- **Subject:** refers to the aspect of the AI system that is being explained. In XAI, this distinction is commonly associated with the difference between local and global explanations. A local explanation describes why an AI system produced a specific output in a given situation. A global explanation aims to describe the general functioning of the AI system.

These three properties of an explanation are entirely determined by the explainability method (Definition 1.1.5) used to produce it. In this context, an explainability method is a transformation process that maps a source of information into an explanation characterized by a specific form and subject. Research in XAI focuses on developing new methods capable of performing this transformation in order to overcome the lack of interpretability associated with DL models.

1.2 Taxonomy of XRL Methods

Now that the concept of XAI has been defined, we turn to its application in RL, known as XRL. As in XAI, explanations in XRL are produced by explainability methods and can be described in terms of their subject, source, and form. This section reviews the possible variations of these three aspects in the specific context of RL.

1.2.1 Subject

The first question to address is the subject of explanation in the context of RL. To answer this question, it is necessary to identify the different components involved in a RL system and determine which of them constitutes the object of explanation. A RL system can be described through two main components: the environment and the policy.

The environment defines the dynamics in which the agent evolves, including the possible states, observations, actions, and transitions. Environments are simulations designed by humans, and their dynamics are explicitly defined. Therefore, it is assumed that the environment is known to the observer and does not require explanation.

173 The main object of interest is therefore the agent’s policy. The policy
 174 represents the decision-making mechanism of the agent. It is a function that maps
 175 observations to actions and is learned through interaction with the environment in
 176 order to maximize the expected cumulative reward. The policy is the component
 177 that encodes the learning process and that XRL aims to explain.

178 Explaining a policy introduces additional challenges compared with classical
 179 XAI settings. In traditional XAI problems, the objective is to explain an isolated
 180 decision, such as determining which features of an image led to a classification
 181 result. In contrast, RL introduces a temporal dimension into the decision-making
 182 process. An action is not only determined by the current observation but also
 183 influences future states and rewards. Therefore, understanding the behavior of
 184 an agent may require considering a sequence of interactions between the agent
 185 and its environment.

186 This temporal dimension leads to a distinction between two objectives of
 187 explanation in XRL:

- 188 • **Decision explanations:** focus on identifying the elements of the current
 189 observation that influenced the action selected by the policy. This type of
 190 explanation is close to classical XAI approaches, as it analyzes the rela-
 191 tionship between inputs and outputs while abstracting away the sequential
 192 nature of the decision process.
- 193 • **Strategy explanations:** aim to characterize the agent’s behavior over
 194 time. They focus on sequences of decisions and interactions with the
 195 environment in order to understand how the agent achieves its long-term
 196 objective. This type of explanation is specific to RL, as it addresses the
 197 temporal and goal-oriented nature of the learning process.

198 These two objectives should not be considered as strictly separated categories,
 199 but rather as two extremes of a continuum. Understanding the elements of an
 200 observation that influence a specific decision can help the observer generalize
 201 this information to infer broader aspects of the agent’s behavior. Conversely,
 202 understanding the agent’s overall strategy can provide insights into the reasoning
 203 behind individual decisions.

1.2.2 Source

We identify three sources of explanation in RL, all obtained during or after the learning process and containing information about the agent:

- **Policies:** policies are functions that generate a distribution over the action space for a given observation. The objective of RL is to learn a policy that maximizes the expected cumulative future rewards. During the learning phase, the policy is iteratively improved through interactions with the environment.
- **Trajectories:** trajectories represent sequences of successive interactions between the agent and the environment. At each step, the agent selects an action based on its observation, which modifies the environment and produces a reward as well as a new observation. This process continues until the end of the trajectory.
- **Value functions:** value functions provide predictions about the future outcome of a trajectory from a given observation. The most commonly used value function is the critic function, which estimates the expected sum of future rewards. This prediction guides the learning process, as the agent updates its policy to maximize the estimated future return. However, other types of value functions can also be used as sources of explainability.

1.2.3 Form

We identify six forms of explanation in the RL setting:

- **Policy Model:** an interpretable model of the agent’s policy function.
- **Saliency Map:** a weighting of the input features representing their influence on the output of the policy function.
- **Counterfactual:** a modified observation obtained by perturbing the original observation in order to produce a different action from the agent.
- **Agent Trajectory:** a sequence of successive interactions between the agent and the environment.
- **Trajectory Prediction:** a prediction about the future evolution of the trajectory produced by a value function.
- **Interaction Model:** an interpretable model of the interactions between the agent and the environment.

Among the three components characterizing an explanation, the form is the one that varies most significantly across explainability methods. While subjects and sources tend to be relatively stable in the XRL literature, the diversity in how explanations are presented is considerably greater. The list above is by no means definitive and may evolve as new approaches emerge. For this reason, the following section is organized according to the form of explanations and presents existing explainability methods based on this criterion.

Explanation Subject	Explanation Form	Explainability Method Family	Source of Explanation		
			Policy	Trajectories	Value functions
Decision	Policy Model	<i>Interpretable policy learning</i>	✓		
		<i>Policy approximation via interpretable model</i>	✓		
	Saliency Map	<i>Observation perturbation</i>	✓		
		<i>Observation gradient analysis</i>	✓		
		<i>Attention-based architecture</i>	✓		
	Counterfactual	<i>Latent space usage for counterfactual generation</i>	✓		
Strategy	Agent Trajectory	<i>Hierarchical decomposition</i>	✓	✓	
		<i>Extraction via critic analysis</i>		✓	✓
	Trajectory Prediction	<i>Critic enrichment</i>			✓
	Interaction Model	<i>Bayesian network learning</i>		✓	
		<i>Latent space usage for MDP generation</i>	✓		

Table 1.1: Taxonomy of explainability methods in reinforcement learning.

1.3 State-of-the-Art Review of XRL Methods

The purpose of this section is to detail the functioning of existing explainability methods (see Table 1.1). They are presented according to the form of explanation they produce, following the taxonomy introduced in Section 1.2.

A family of methods is defined as a set of methods that share the same source, form, and subject of explanation, as well as a similar procedure for generating explanations. Methods belonging to the same family may differ in their implementation details while relying on the same principle.

The remainder of this review is organized by form of explanation, following the six forms identified in Section 1.2.3. For each form, we first describe its principle and its limitations, then present the families of methods that produce it.

1.3.1 Policy Model

The policy function can be represented by a wide variety of function classes. In modern RL, NN is the most commonly used models for representing policies. Although these models achieve the highest performance, their connectionist nature makes their internal representations non-interpretable.

For this reason, a substitutive approach to XRL aims to replace these models with alternative models considered interpretable. Such models can be understood through direct analysis of their representations. The main classes of interpretable models used to represent policy functions include decision trees [Topin et al., 2021], algorithms [Trivedi et al., 2022], formal logic [Payani and Fekri, 2019], and linear models [Garreau and von Luxburg, 2020].

Explanations derived from an interpretable model primarily provide insights into the decision-making process, since what is modeled is the mapping between an observation and an action. If the model is sufficiently well structured, the

observer can form a mental representation of a sequence of decisions across the environment and thus obtain information about the agent’s strategy.

The main limitation of these methods is that interpretable models rely on symbolic representations. When the policy to be represented is too complex, achieving comparable performance requires a large number of rules or conditions. The resulting model can quickly become difficult to analyze and lose its interpretability.

For example, representing a policy capable of playing Go using a symbolic model would require considering a very large number of possible situations and specific cases. Even if such a representation were possible, the resulting model would likely become intractable for human analysis due to its complexity. This illustrates why NN-based approaches have become dominant in such domains, as they can learn complex decision strategies without requiring an explicit enumeration of rules.

To obtain an interpretable policy model, two families of explainability methods are available: *interpretable policy learning* and *policy approximation via interpretable model*.

Interpretable Policy Learning

This family of methods aims at learning a policy represented by an interpretable model [Topin et al., 2021, Trivedi et al., 2022]. It is the most direct approach to addressing the challenges posed by XRL. This type of training requires modifying the Markov Decision Processes (MDP) in order to adapt it to the search space and to the type of policy representation. Moreover, this search space often needs to be reduced to make learning feasible. Such a reduction requires the integration of expert knowledge into the learning process. With this family of methods, the *interpretable policy function* itself is used to provide explanations to the observer.

Policy Approximation via Interpretable Model

An interpretable model can be trained to approximate a policy learned through deep RL [Sieusahai and Guzdial, 2021]. In this setting, the problem is formulated as a supervised learning task, where the original deep RL policy is used to label each observation with the action it selects, such as the observation from the Atari game “Pong” shown in Figure 1.3a. Using this dataset, the interpretable model is trained to reproduce the behavior of the original policy, yielding for instance the decision tree illustrated in Figure 1.3b, obtained by approximating a deep RL policy trained on that environment. The resulting interpretable model is then used as an explanation for the observer. If this approximation is sufficiently faithful, it can also be deployed in the environment in place of the original policy, thereby providing a high degree of control over the agent.

1.3.2 Saliency Map

The influence of each part of the input on the action selected by the policy can be visualized using heatmaps (Figure 1.4). In the RL setting, the input corresponds to the observation received by the agent at a given time step. Heatmaps therefore highlight the regions of the observation that have the greatest influence on the

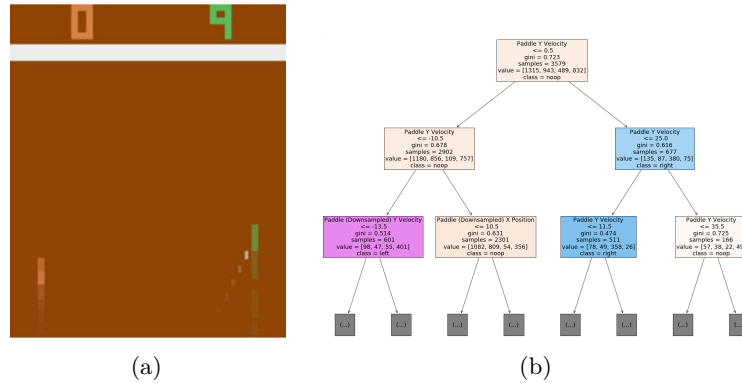


Figure 1.3: Policy approximation via an interpretable model [Siesahai and Guzdial, 2021].

action selection process. By providing this information, they allow the observer to identify which elements of the environment are considered important by the policy when making a decision. In Figure 1.4, both panels correspond to the same observation of the Atari game “Breakout”, with panel (a) obtained by observation perturbation and panel (b) by observation gradient analysis.

A limitation of heatmap-based methods is that the resulting explanations are not always directly meaningful to the observer. As illustrated in Figure 1.4b, interpreting which regions are important often requires human reasoning and involves a degree of subjectivity. This limitation becomes particularly relevant in RL, where the objective is not only to explain a single decision but also to understand the factors influencing a sequence of decisions over time.

Three families of explainability methods can be used to obtain this type of explanation: *observation perturbation*, *observation gradient analysis*, and *attention-based architectures*.

Observation Perturbation

Perturbation analysis consists in applying various modifications to the observation in order to study their impact on the output of the policy function [Greydanus et al., 2018]. The greater the variation in the output caused by a perturbation applied to a given component of the input vector, the higher the weight assigned to that component. The magnitude of the perturbation is measured by computing the distance between the original output and the output obtained from the perturbed input. Using this set of weights, a *heatmap* is constructed, enabling the observer to visualize which parts of the observation vector are most sensitive in the action-selection process (see Figure 1.4a).

Observation Gradient Analysis

Observation gradient analysis refers to a set of methods that require a differentiable policy function in order to extract information from it [Simonyan et al., 2014, Weitekamp et al., 2019, Huber et al., 2019]. To this end, the gradient of the policy function is computed with respect to all components of the input vector. For each input component, the resulting gradient provides a weighting that

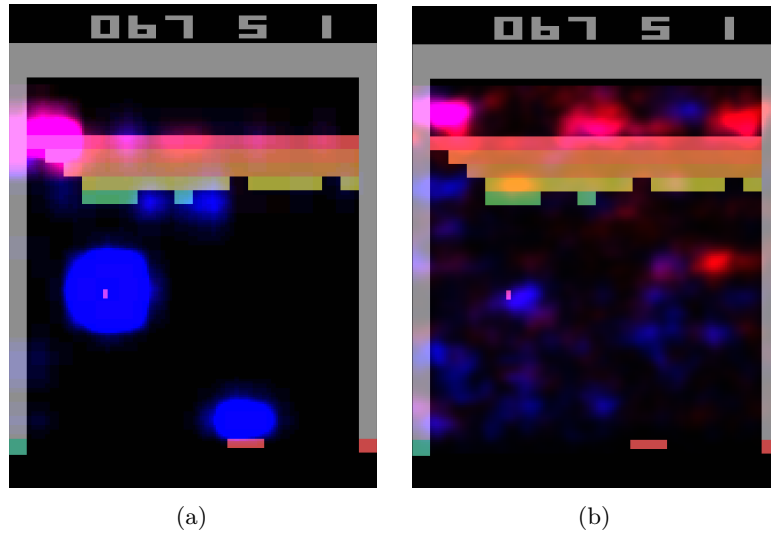


Figure 1.4: Heatmaps for the same observation of the Atari game “Breakout” [Greydanus et al., 2018].

343 represents its importance in the output of the policy function. Based on these
 344 weightings, a *heatmap* is constructed to allow the observer to visually identify
 345 the parts of the input vector that are the most influential in the policy function’s
 346 output (see Figure 1.4b).

347 *Attention-Based Architecture*

348 Attention blocks constitute a NN architecture that makes it possible to weight
 349 the relative importance of one part of a vector with respect to the other parts
 350 of the same vector [Vaswani et al., 2023]. These weights are computed as a
 351 function of the vector itself and are learned by the NN during training. For
 352 a given observation, each component of the observation vector is assigned a
 353 scalar value. It can therefore be assumed that the components associated
 354 with the highest weights are the most influential in determining the output
 355 [Mott et al., 2019, Itaya et al., 2021]. This attention-based *heatmap* is then
 356 provided to the observer as the explanation, as illustrated for the Atari game
 357 “Breakout” in Figure 1.5a, where the highlighted regions are easily interpretable.
 358 However, as shown for “Ms. Pac-Man” in Figure 1.5b, the attention weights are
 359 not always concentrated on a small, easily interpretable set of regions, which
 360 can make this type of explanation difficult to exploit.

361 1.3.3 Counterfactual

362 A counterfactual explanation is a new observation, close to the original one, that
 363 leads the agent to select a different action. Comparing the two observations
 364 reveals which elements of the environment must change for the agent to behave
 365 differently, and thus provides insight into its decision-making process. Unlike
 366 the other forms of explanation, this form is currently associated with a single

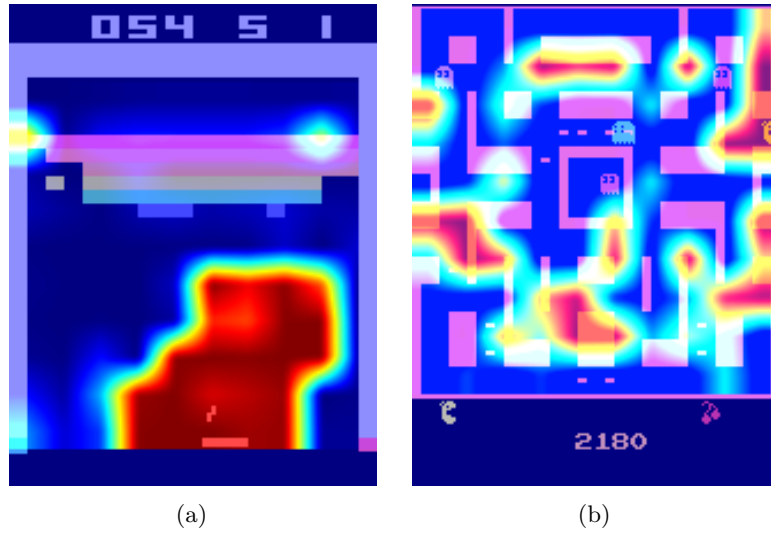


Figure 1.5: Attention-based heatmaps for the Atari games “Breakout” and “Ms. Pac-Man” [Mott et al., 2019].

family of methods, based on the *use of the Latent Space (LS) for counterfactual generation*.

Latent Space Usage for Counterfactual Generation

This family of methods uses generative models to build counterfactuals from the Latent Representation (LR) of the observation [Bond-Taylor et al., 2021, Olson et al., 2021]. The observation is encoded into the LS, its representation is modified until the agent selects the desired action, and this modified representation is decoded back into a new observation. Working in the LS rather than at the pixel level keeps the counterfactual close to the original state and consistent with the environment. The fidelity of the generated observations to the real game can then be assessed through user studies [Olson et al., 2021].

Figure 1.6 illustrates this approach on the Atari game *Space Invaders* [Olson et al., 2021]. In the original observation (Figure 1.6a) the agent selects “LEFT”, whereas in the counterfactual observation (Figure 1.6b) it selects “RIGHT_FIRE”. Here, removing an enemy is enough to change the action, which suggests that the position of that enemy influences the agent’s decision. Such observations remain only indicative. Identifying a reliable pattern requires repeating the analysis across many situations.

1.3.4 Agent Trajectory

Until now, the presented methods have focused on explaining individual decisions made by the agent. The methods considered from this point on aim to provide insight into the agent’s strategy by analyzing its behavior over time.

The most direct approach consists in visualizing agent trajectories, most commonly through complete episodes. By analyzing these trajectories, the observer can infer the dynamics underlying the agent’s behavior and how its

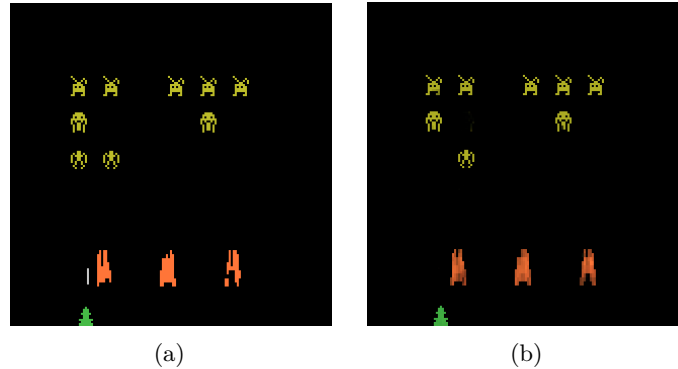


Figure 1.6: Counterfactual explanation of the agent’s behavior for the Atari game *Space Invaders* [Olson et al., 2021]. Panel a shows the original observation and panel b the counterfactual observation produced by a generative model.

decisions evolve through time. This type of visualization is not specific to XRL. In practice, visualizing agent episodes is a standard step in the development and evaluation of RL. It provides a simple way to verify that the agent behaves as expected and to detect potential implementation errors.

This simple approach can be further enhanced using families of methods such as *hierarchical decomposition* and *extraction via critic analysis*.

Extraction via Critic Analysis

In this family of methods, the predictions of the critic are used to determine which trajectories to present to the observer [Sequeira and Gervasio, 2020]. A set of states is selected from multiple trajectories. For a given state, all possible actions are examined. The state is assigned a value equal to the difference between the critic’s prediction for the lowest-valued action and the prediction for the highest-valued action. Trajectories containing the states with the highest values according to this procedure are then chosen to be presented to the observer. These trajectories are considered critical moments of the agent and are therefore relevant for analysis. For this family of methods, the *selected trajectories* represent the interpretable information provided to the observer.

Hierarchical Decomposition

Hierarchical agents are composed of a set of sub-policies specialized for specific tasks within subsets of the environment’s states. This family of methods distributes the optimization of the reward function across multiple models, significantly simplifying the learning process. Moreover, this decomposition of the policy function can also provide explainability in the agent’s strategy [Beyret et al., 2019]. If a sub-policy is selected, it indicates that the agent will execute a specialized behavior. This behavioral specialization provides insight into the strategy the agent employs to maximize the reward function. The agent’s *trajectory*, together with the *sub-policy* selected for each action, is used as interpretable information.

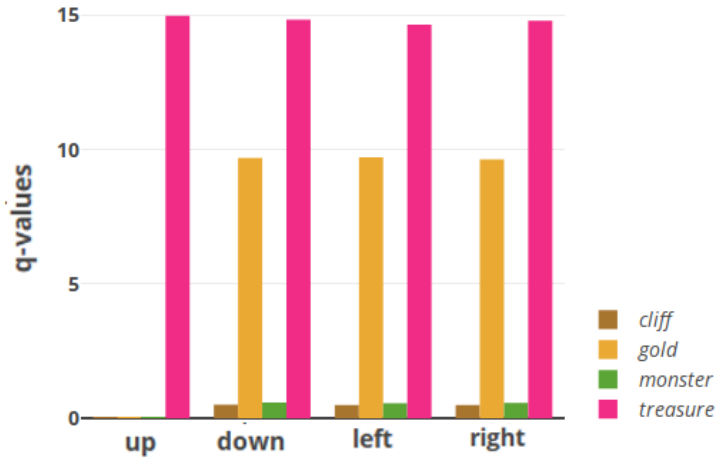


Figure 1.7: Prediction of sub-rewards as a function of the action selected by the agent for a given state [Juozapaitis et al., 2019].

1.3.5 Trajectory Prediction

Making predictions about the future of a trajectory allows one to form a representation of what the system expects, based on its training experience. These predictions are produced using value functions. Such functions provide insights into the future evolution of the agent’s trajectory. By their nature, this form of explanation informs the observer about the strategy adopted by the agent. Value functions provide an estimate of the expected cumulative reward from a given state. Consequently, a value function maps an observation to a scalar value in $\mathbb{R}$.

This aggregated prediction provides limited information about the underlying factors that contribute to this value. It does not directly reveal which aspects of the task are being optimized or how the predicted return is composed. Unlike the other forms of explanation, this form is currently associated with a single family of methods, aimed at improving the interpretability of value-based explanations: *critic enrichment*.

Critic Enrichment

In order to obtain interpretable information from the predictions of the critic, it is possible to decompose the critic into multiple sub-functions [Juozapaitis et al., 2019]. The reward function is thus decomposed into multiple sub-reward functions, each representing a sub-objective of the task to be accomplished. Similarly, for each of these sub-reward functions, corresponding sub-critic functions are defined, along with a function that combines their outputs. During training, the agent selects its actions so as to maximize this combination function of the sub-critics. Using this approach, for each observation and action, a prediction is obtained regarding the sub-objectives the agent is attempting to maximize, so that the action chosen by the agent reveals which sub-reward it aims to maximize (see Figure 1.7). This *prediction* is then used as the explanation for the observer.

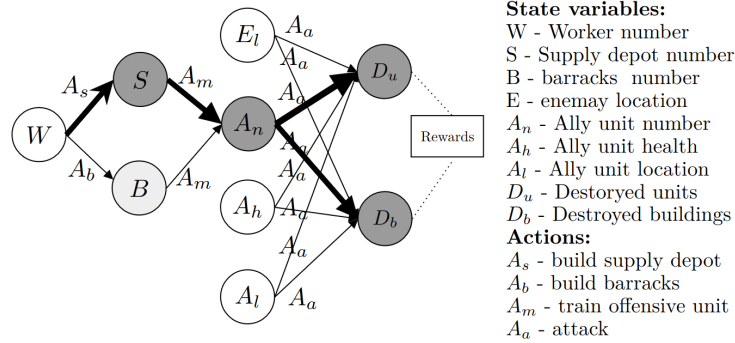


Figure 1.8: Agent-environment interaction model for the video game *StarCraft II* [Madumal et al., 2020].

1.3.6 Interaction Model

Modeling the interactions between an agent and its environment makes it possible to understand how the agent's decisions are made and what impact they have on the environment. By incorporating these interactions into an explanatory model, one can better grasp the complex relationships between the agent and its environment, thereby identifying the determining factors in the decision-making process. By explicitly handling the agent-environment interaction, this type of explanation aims to explain the agent's strategy. The families of methods used to obtain explanations in the form of agent-environment interactions are: *Bayesian network learning* and *LS usage for MDP generation*.

Bayesian Network Learning

The evolution of an agent within an environment can be observed as a set of random variables connected through a Bayesian network [Madumal et al., 2020]. Random variables are defined to represent both external aspects (the environment) and internal aspects of the agent (the actions). By analyzing trajectories, it becomes possible to infer causal relationships among these random variables. At the end of this process, a weighted causal graph is obtained, which serves as an explanation for the observer, as illustrated for the video game *StarCraft II* in Figure 1.8, where nodes represent random variables and edges represent causal relationships between them. This *Bayesian network*, representing the interactions between the agent and the environment, is then used as the explanation.

Latent Space Usage for MDP Generation

Each layer of a NN can be viewed as a projection from one space to another [Zahavy et al., 2017]. As information propagates through successive layers of a NN, it becomes increasingly abstract. It can therefore be assumed that two vectors that are close in this projected space are also close in terms of the agent's perception. Based on this observation, it is possible to redefine an MDP by leveraging the abstract representations produced by the final layers of a NN. Once the new MDP has been constructed, a model of the interaction between

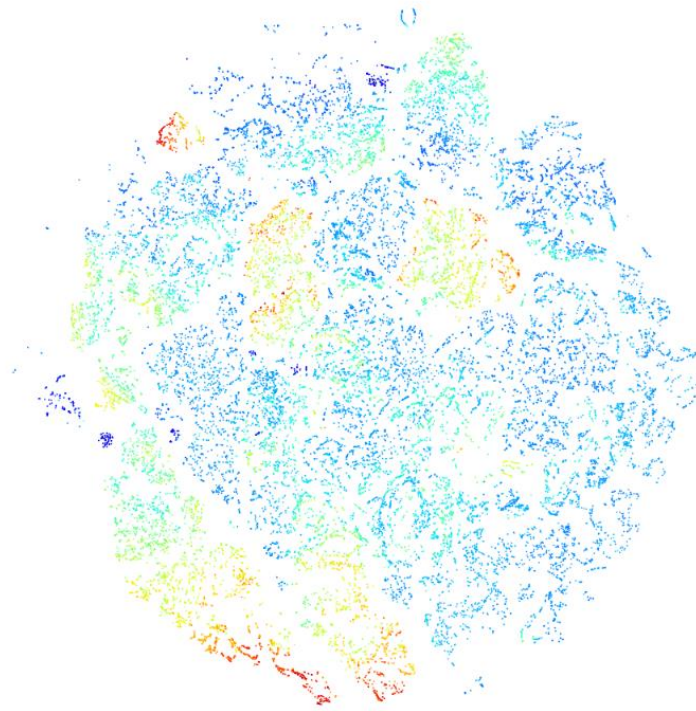


Figure 1.9: t -SNE projection of a neural network's latent space [Zahavy et al., 2017].

the agent and its environment is obtained. This model, represented as a *graph*, is then used as the explanation.

Figure 1.9 illustrates this idea using a two-dimensional t -SNE projection of the LS of the NN used to represent the policy. Each point is colored according to the value function's prediction for the corresponding state, with warmer colors indicating higher predicted values. An agreement can be observed between the structure of this projected space and the critic's predictions: states that are assigned similar values by the critic also tend to be located close to one another in the projection. This agreement makes it possible to manually identify clusters of states, which likely correspond to states that are perceived as similar by the agent.

1.4 Conclusion

The taxonomy and the state-of-the-art review presented in this chapter make it possible to draw three main observations.

1.4.1 Absence of Explanation of the Learning Process

Regarding the subject of explanation, Table 1.1 shows that none of the reviewed methods addresses the learning process itself. Such an explanation would concern

493 how the policy is progressively updated throughout training, and why it evolves
 494 over successive iterations to converge toward a particular solution. Several
 495 reasons explain this absence.

496 First, among the different machine learning paradigms, RL is arguably the
 497 one that would benefit the most from explanations of the learning process.
 498 Unlike supervised or unsupervised learning, it relies on an explicit exploration–
 499 exploitation trade-off, making the learning trajectory itself an informative subject
 500 of explanation. However, most XRL methods are inherited from the XAI
 501 literature, where explanations focus exclusively on the behavior of already-
 502 trained models. This limitation has naturally carried over to XRL.

503 Second, from a practical perspective, explanations of the learning process
 504 have limited interest. In most real-world applications, practitioners deploy an
 505 already-trained agent, while training itself takes place offline. Users are generally
 506 interested in understanding the behavior of the final policy rather than the
 507 optimization process that produced it.

508 Finally, explaining the learning process raises significant methodological
 509 challenges. Training in RL can be viewed as the iterative improvement of a policy
 510 through repeated interactions with the environment. In this sense, explaining
 511 learning would amount to explaining the evolution of an entire sequence of
 512 policies rather than a single one. Current XRL methods still struggle to provide
 513 comprehensive explanations of an individual policy, so extending them to a full
 514 training trajectory appears considerably more difficult.

515 1.4.2 Lack of Unified Metrics for Explainability

516 The diversity of explanation forms identified in this review also reveals another
 517 limitation shared by nearly all families of XRL methods, which is the absence
 518 of reliable metrics for evaluating explainability. In many studies, explanations
 519 are not evaluated with human participants. Instead, authors focus on technical
 520 properties of the proposed method, such as fidelity, sparsity, or computational
 521 efficiency. When human evaluations are conducted, they usually rely on subjective
 522 satisfaction questionnaires. Although useful, self-reported satisfaction remains
 523 prone to numerous biases and does not necessarily indicate that the explanation
 524 has improved the observer’s mental model or understanding of the agent.

525 This limitation is not specific to XRL, it reflects a broader challenge affecting
 526 XAI as a whole. In our view, this limitation comes from how explainability has
 527 developed in artificial intelligence. Most of the proposed approaches have been
 528 developed by the computer science community. Explainability has consequently
 529 often been studied from an algorithmic or mathematical perspective, with less
 530 attention given to how humans actually benefit from explanations.

531 Our definition of explanation explicitly places the observer at the center of the
 532 explanatory process. From this perspective, the value of an explanation should
 533 not only depend on its technical properties but also on its ability to improve
 534 the observer’s understanding. Addressing this challenge will require stronger
 535 interdisciplinary collaboration between computer science and disciplines such
 536 as cognitive science, psychology, philosophy, and sociology. These fields provide
 537 the theoretical and experimental tools needed to study human understanding
 538 and to develop evaluation protocols that measure the actual effectiveness of
 539 explanations.

540 1.4.3 Toward Intrinsic Explainability in XRL

541 Our review also suggests that current XRL research remains largely dominated
542 by substitutive explainability approaches. Replacing a deep model with a
543 simpler one therefore risks losing precisely the complexity that makes deep RL
544 effective. For this reason, we argue that a promising direction for XRL is to
545 develop explanations directly from the NN that forms the agent’s policy. Such
546 approaches aim to analyze the representations learned by the network itself.
547 From this perspective, the NN should not necessarily be regarded as an black
548 box, but as a complex representation whose apparent opacity may instead reflect
549 the limitations of current analysis tools.

550 Among the methods reviewed, *latent space usage for MDP generation* (Sec-
551 tion 1.3.6) appears closest to this intrinsic perspective. However, this approach
552 still relies on a largely manual and qualitative analysis of the LS, which limits
553 its reproducibility and its ability to produce systematic explanations. The next
554 chapter builds upon this idea and proposes a mechanistic XRL approach based
555 on LS analysis.

556 Acronyms

557 **AI** Artificial Intelligence 1, 5, 6

558 **DL** Deep Learning 1, 5, 6

559 **LLM** Large Language Model 1

560 **LR** Latent Representation 13

561 **LS** Latent Space 13, 16, 17, 19

562 **MDP** Markov Decision Processes 10, 16

563 **NN** Neural Network 5, 9, 10, 12, 16, 17, 19

564 **RL** Reinforcement Learning 1, 2, 6–11, 14, 18, 19

565 **XAI** eXplainable Artificial Intelligence 1, 2, 5–7, 18

566 **XRL** eXplainable Reinforcement Learning 1, 2, 6–10, 14, 18, 19

Bibliography

- [Alaarabiou et al., 2024] Alaarabiou, M., Delestre, N., and Vercouter, L. (2024). Explicabilité en Apprentissage par Renforcement : vers une Taxinomie Unifiée. In 18èmes Journées d’Intelligence Artificielle Fondamentale, La rochelle, France.
- [Barredo Arrieta et al., 2020] Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-López, S., Molina, D., Benjamins, R., Chatila, R., and Herrera, F. (2020). Explainable artificial intelligence (xai): Concepts, taxonomies, opportunities and challenges toward responsible ai. Information Fusion, 58:82–115. **Article**.
- [Bellucci et al., 2021] Bellucci, M., Delestre, N., Malandain, N., and Zanni-Merk, C. (2021). Towards a terminology for a fully contextualized xai. Procedia Computer Science, 192:241–250. **Article**.
- [Beyret et al., 2019] Beyret, B., Shafti, A., and Faisal, A. A. (2019). Dot-to-dot: Explainable hierarchical reinforcement learning for robotic manipulation. In 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE. **Article**.
- [Bond-Taylor et al., 2021] Bond-Taylor, S., Leach, A., Long, Y., and Willcocks, C. G. (2021). Deep generative modelling: A comparative review of vaes, gans, normalizing flows, energy-based and autoregressive models. **Article**.
- [Chen et al., 2021] Chen, M., Tworek, J., Jun, H., Yuan, Q., de Oliveira Pinto, H. P., Kaplan, J., et al. (2021). Evaluating large language models trained on code. **Article**.
- [Doshi-Velez and Kim, 2017] Doshi-Velez, F. and Kim, B. (2017). Towards a rigorous science of interpretable machine learning. **Article**.
- [Garreau and von Luxburg, 2020] Garreau, D. and von Luxburg, U. (2020). Explaining the explainer: A first theoretical analysis of lime. In Chiappa, S. and Calandra, R., editors, Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics, volume 108 of Proceedings of Machine Learning Research, pages 1287–1296. PMLR. **Article**.
- [Greydanus et al., 2018] Greydanus, S., Koul, A., Dodge, J., and Fern, A. (2018). Visualizing and understanding atari agents. **Article**. **Code**.
- [Gunning and Aha, 2019] Gunning, D. and Aha, D. (2019). Darpa’s explainable ai (xai) program. AI Magazine, 40(2):44–58. **Article**.

- [Huber et al., 2019] Huber, T., Schiller, D., and André, E. (2019). Enhancing explainability of deep reinforcement learning through selective layer-wise relevance propagation. In *KI 2019: Advances in Artificial Intelligence: 42nd German Conference on AI, Kassel, Germany, September 23–26, 2019, Proceedings 42*, pages 188–202. Springer. **Article**.
- [Itaya et al., 2021] Itaya, H., Hirakawa, T., Yamashita, T., Fujiyoshi, H., and Sugiura, K. (2021). Visual explanation using attention mechanism in actor-critic-based deep reinforcement learning. **Article**.
- [Juozapaitis et al., 2019] Juozapaitis, Z., Koul, A., Fern, A., Erwig, M., and Doshi-Velez, F. (2019). Explainable reinforcement learning via reward decomposition. In *IJCAI/ECAI Workshop on explainable artificial intelligence*. **Article**.
- [Lewis et al., 2021] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., tau Yih, W., Rocktäschel, T., Riedel, S., and Kiela, D. (2021). Retrieval-augmented generation for knowledge-intensive nlp tasks.
- [Madumal et al., 2020] Madumal, P., Miller, T., Sonenberg, L., and Vetere, F. (2020). Explainable reinforcement learning through a causal lens. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 2493–2500. **Article**.
- [Miller, 2019] Miller, T. (2019). Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence*, 267:1–38. **Article**.
- [Mnih et al., 2013] Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., Wierstra, D., and Riedmiller, M. (2013). Playing atari with deep reinforcement learning.
- [Mott et al., 2019] Mott, A., Zoran, D., Chrzanowski, M., Wierstra, D., and Jimenez Rezende, D. (2019). Towards interpretable reinforcement learning using attention augmented agents. *Advances in neural information processing systems*, 32. **Article**.
- [Olah et al., 2020] Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., and Carter, S. (2020). Zoom in: An introduction to circuits. *Distill*. **Article**.
- [Olson et al., 2021] Olson, M. L., Khanna, R., Neal, L., Li, F., and Wong, W.-K. (2021). Counterfactual state explanations for reinforcement learning agents via generative deep learning. *Artificial Intelligence*, 295:103455. **Article**. **Code**.
- [Payani and Fekri, 2019] Payani, A. and Fekri, F. (2019). Inductive logic programming via differentiable deep neural logic networks. **Article**. **Code**.
- [Rudin, 2019] Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215. **Article**.
- [Sequeira and Gervasio, 2020] Sequeira, P. and Gervasio, M. (2020). Interestingness elements for explainable reinforcement learning: Understanding agents’ capabilities and limitations. *Artificial Intelligence*, 288:103367. **Article**.

- [Seshia et al., 2022] Seshia, S. A., Sadigh, D., and Sastry, S. S. (2022). Toward verified artificial intelligence. *Communications of the ACM*, 65(7):46–55. **Article.**
- [Siegel and Craver, 2024] Siegel, G. and Craver, C. F. (2024). Phenomenological laws and mechanistic explanations. *Philosophy of Science*, 91(1):132–150.
- [Sieusahai and Guzdial, 2021] Sieusahai, A. and Guzdial, M. (2021). Explaining deep reinforcement learning agents in the atari domain through a surrogate model. **Article.**
- [Silver et al., 2017] Silver, D., Hubert, T., Schrittwieser, J., Antonoglou, I., Lai, M., Guez, A., Lanctot, M., Sifre, L., Kumaran, D., Graepel, T., et al. (2017). Mastering chess and shogi by self-play with a general reinforcement learning algorithm. *arXiv preprint arXiv:1712.01815*.
- [Simonyan et al., 2014] Simonyan, K., Vedaldi, A., and Zisserman, A. (2014). Deep inside convolutional networks: Visualising image classification models and saliency maps. **Article. Code.**
- [Topin et al., 2021] Topin, N., Milani, S., Fang, F., and Veloso, M. (2021). Iterative bounding mdps: Learning interpretable policies via non-interpretable methods. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 9923–9931. **Article.**
- [Trivedi et al., 2022] Trivedi, D., Zhang, J., Sun, S.-H., and Lim, J. J. (2022). Learning to synthesize programs as interpretable and generalizable policies. **Article. Code.**
- [Vaswani et al., 2023] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. (2023). Attention is all you need. **Article.**
- [Weitkamp et al., 2019] Weitkamp, L., van der Pol, E., and Akata, Z. (2019). Visual rationalizations in deep reinforcement learning for atari games. **Article.**
- [Zahavy et al., 2017] Zahavy, T., Zrihem, N. B., and Mannor, S. (2017). Graying the black box: Understanding dqns. **Article.**
- [Zednik, 2019] Zednik, C. (2019). Solving the black box problem: A normative framework for explainable artificial intelligence. *arXiv preprint arXiv:1903.04361*.